TAKE-HOME · EVALUATION TOOL FOR CODING-AGENT HARNESSES

harness-eval · one page

Answers one question before a harness change reaches every engineer: **is the coding agent producing better code after this change?** Evidence first: repeated paired runs, hidden tests, and a verdict that is allowed to say "not proven".

5 real tasks | **60** Claude Code runs | **$5.28** total spend | **4** agents: Claude · Codex · CrewAI/Ollama · replay | **18** pipeline tests, $0

CONCEPTS I DEFINED, AND WHY

Harness	A directory: `harness.yaml` + files overlaid on the repo + the agent's private home. One shape for AGENTS.md, skills, MCP, hooks, model or agent swaps.
Suite	Real business requests + **hidden acceptance tests**, injected only after the agent exits.
4 dimensions	**Works** and **intent** are guardrails that can veto a change. **Conventions** (AST rules from AGENTS.md) and **cost** (USD, tokens, time) cannot.
Paired delta	Unit = task, so difficulty cancels. A bootstrap over tasks *and* trials gives a 95% CI.
Noise floor	`calibrate` = baseline vs itself (A/A). It must say INCONCLUSIVE; its spread is the bar real effects must clear.
Verdict	IMPROVEMENT 0 · REGRESSION 1 · INCONCLUSIVE 2 · TRADE-OFF 3. Exit codes gate CI; TRADE-OFF means no invented weights.

THE FIVE HARDEST DECISIONS

1. **No LLM judge.** Tests and AST rules only. A judge covers more but reintroduces "an opinion formed by reading outputs", the thing the brief wants replaced.
2. **Failed runs score zero.** Timeouts and crashes are harness behaviour; dropping them flatters a flaky harness.
3. **Guardrails veto, no weighted score.** Any detected drop in works or intent = REGRESSION; mixed results go to a human.
4. **Isolation from my own harness.** A hand check caught my global `CLAUDE.md` leaking in through parent-directory lookup. Now: runs outside the profile, private HOME/config, ancestor scan, per-run plugin/MCP check.
5. **A/A floor on top of the CI.** Deliberately conservative: a false "ship it" reaches every engineer, a missed gain costs one more run.

WHAT I CUT, AND WHY

LLM judge: opinion, not evidence

Parallel runs: 1 GPU, rate limits; arms interleaved

Full Codex demo: adapter + isolation spike only, budget

Non-Python rules: one stack done well

Tasks mined from real PRs: 5 hand-written

Power analysis: "how many trials?" **History dashboards**

ONE RESULT I DID NOT TRUST

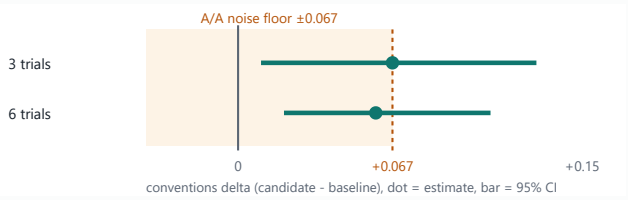
INCONCLUSIVE

baseline: bare AGENTS.md → candidate: + team conventions

Claude Code on Haiku, 5 tasks × 3 trials. The tool said **not proven**, but the evidence underneath was not empty:

signal (per run)	baseline	candidate
Team conventions score	0.84	**0.90**
Runs that added no tests	28 / 30	**18 / 30**
Raised a built-in exception	1 / 30	**0 / 30**
Cost per task	$0.083	$0.093

The 3-trial verdict came down to a **tie with the noise floor** (+0.067 vs 0.067).



> **What I did:** I did not relax the rule after seeing the data; that is how a tool learns to say what you want. I bought more data instead. Calibration had already cached baseline trials 3–5, so 6 trials cost 15 runs ($1.39).

At 6 trials: still INCONCLUSIVE. +0.06, CI [0.02, 0.11]: real but small, under the floor. The new "add tests" rule is obeyed only 40% of the time; cost +12% (not significant). The only hidden-test failure in 60 runs was a candidate run (coupon rounding).

My call: don't roll it out on this evidence. Make the tests rule enforceable (a pytest hook), then re-run.

> **Also distrusted:** the baseline passed *every* hidden test (intent 1.00, floor 0). This suite cannot catch a small hit to correctness. The next investment is harder tasks, not more statistics.

Reproduce for $0: `uv run harness-eval evaluate examples/orders … --offline`

Details: README.md · docs/ · process/